Supplementary Information for
An attractor account of numerosity perception

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This document accompanies the article *An attractor account of numerosity perception*. Supplementary Material 1 details the trial-level fitting; Supplementary Material 2 reports the full model comparison and the constrained cross-duration fits; Supplementary Material 3 the parameter recovery; Supplementary Material 4 the resolution-limit prediction and its robustness; and Supplementary Material 5 the exploratory developmental-dyscalculia reanalysis.

Supplementary Material 1: Trial-level fitting

Data. We reconstruct one row per trial from the openly available data of Cheyette and Piantadosi (2020), grouping by participant and trial and keeping the presented numerosity, the typed response, and the response time (converted to seconds). We use the five DURATION conditions (40, 80, 160, 320, 640 ms), true numerosities $n = 1$–15, giving ≈10,600 trials per duration and ≈53,000 in total. No trials are aggregated: the likelihood is evaluated over the individual integer reports.

Response likelihood. For each model and duration the response probability of Eq. (4) marginalises the readout over the sensory channel $y_0 \sim \mathcal{N}(n, \sigma(n))$. The integral is computed *deterministically*, so the objective is free of Monte-Carlo noise. Gauss–Hermite quadrature is inappropriate here: for the attractor models the clean-up map $y_0 \mapsto x^*$ is near-discontinuous (evidence snaps to basins), and Gauss–Hermite carries a 1.7–4% error even at 241 nodes. We therefore use a uniform standard-normal grid with Gaussian weights for *every* model, so the comparison is on identical footing; validated against high-sample Monte-Carlo, a grid of 801 nodes matches to $\leq 2 \times 10^{-3}$ (mean 1.5×10^{-4}). The optimiser uses a cheaper grid (201 nodes); the reported log-likelihoods are recomputed at the fitted parameters on a fine grid (1601 nodes).

Lapse channel. Typed responses run up to 99, and a few per cent of guesses fall outside the integer-well support (3.9–6.1% across durations). We never discard them: the final response probability is $p_{\text{final}}(r \mid n) = (1 - \ell)\, p(r \mid n) + \ell \,/\, R_{\max}$ with $R_{\max} = 99$, a uniform contamination that places positive mass everywhere. The lapse ℓ is fitted per duration under an identical convention across all models and is counted in AIC/BIC; it is stable across models (≈0.06–0.08).

Response-time and joint channels. Response time is modelled as log-normal, $\log \mathrm{rt} \sim \mathcal{N}(\log(t_0 + m(n)), \sigma_{\mathrm{rt}})$, with the task-fixed base $t_0 = 1.30$ s. The metric $m(n)$ is the expected posterior entropy ($\kappa\, \mathbb{E}[H]$, available to every model), the expected convergence time ($\tau\, \mathbb{E}[T]$, attractor only), or their sum (the hybrid; Eq. 3). Because the log-normal maximum-likelihood fit targets the mean of $\log \mathrm{rt}$, i.e. the geometric-mean RT, all reported R^2 values compare the prediction against the geometric mean rather than the median. The coupled joint $p(r, \mathrm{rt} \mid n)$ integrates response and RT over the same latent nodes; co-optimising all channels together (warm-started from the two-stage fit) leaves the model ranking unchanged and moves the landscape negligibly, so RT can be treated as a downstream cross-prediction.

Optimisation. The deterministic objective is minimised by `scipy` differential evolution (Storn and Price 1997) followed by a Nelder–Mead polish, with fixed seeds. The attractor models require a larger optimiser budget than the smooth static readouts (a low-budget fit understates them); re-fitting at a higher budget with a different seed changes the reported log-likelihoods by less than one part in 10^3, confirming the fits are converged and the model ordering is not an optimisation artefact.

Supplementary Material 2: Model comparison and constrained fits

Table S1 gives the full trial-level response comparison. Every model is fitted under the identical convention (per-duration parameters plus a per-duration lapse, summed over the five durations). The super-linear-width Gaussian attractor is the best account; the ranking isolates the two ingredients that matter.

Clean-up at equal parameters. Turning the clean-up dynamics on is a zero-parameter change (the dynamics are deterministic given the landscape), so the attractor and its

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Table S1. Trial-level response model comparison. Categorical information criteria over the $\sim$53,000 individual responses, summed over the five exposure durations, relative to the best model. k is the total free-parameter count (per-duration mechanism parameters plus lapse). "Attractor" includes the clean-up dynamics; "readout" is the matched static control at the same k. The width-model rows carry one parameter more than the Weber/ANS and normative benchmarks (the free width exponent); at this sample size BIC charges $\approx$9 per parameter against AIC's 2, yet the ranking and margins barely change, so the advantage is not a parameter-count artefact.

Model	k	ΔAIC	ΔBIC
Super-linear attractor (free β)	**25**	**0**	**0**
Super-linear attractor (power-law n^p)	25	165	165
Super-linear readout (free β, static)	25	1256	1256
Power-law readout (static)	25	1491	1491
Weber/ANS ($\sigma_n \propto n$)	20	4435	4398
Normative log-prior proxy	20	7666	7629

Table S2. Fitted parameters of the winning model across exposure durations. σ_c and β set the well-width law $\sigma_n = (\sigma_c/f_n)^\beta$; s_0 and λ_0 the sensory-noise profile $\sigma(n)$; ℓ the lapse. The 40 ms flash is noise-dominated (large s_0, extreme β); from 80 ms the parameters vary smoothly.

Duration	σ_c	s_0	λ_0	β	ℓ
40 ms	0.085	0.023	2.91	3.42	0.080
80 ms	0.054	0.0056	3.03	1.81	0.064
160 ms	0.047	0.0023	3.25	2.02	0.059
320 ms	0.045	0.00047	3.76	2.19	0.079
640 ms	0.041	0.00036	3.67	2.15	0.072

static twin occupy the identical parameter space. Within the super-linear family this flips the fit by $\Delta\text{AIC} = 1256$ in favour of clean-up, and within the power-law family by 1326, showing that the dynamics do explanatory work rather than add flexibility. (Per duration, clean-up helps strongly at 40–160 ms and is roughly neutral at 320–640 ms, where the attractor has largely settled during the exposure itself.)

Width scaling. The width exponent is a free parameter of the model rather than a fixed value. It is estimated firmly super-linear at every exposure except the noise-dominated 40 ms flash (Table S2), which gives the wells their sharp-to-broad profile: narrow and well-separated for small numbers, wide and overlapping for large ones. Granting the Weber/ANS code the same flexibility does not close the gap: freeing its width exponent (the power-law readout, also $k = 25$) leaves it 1491 behind the winning attractor, and the freed exponent settles super-linear rather than at the Weber value $\sigma_n \propto n$, so at equal parameter count a magnitude code still loses and does not prefer the Weber scaling.

Is the landscape stable across durations? The main text fits each duration independently and reads the resulting parameter trajectories; the substantive finding is that the landscape sharpens with exposure (n_{crit} grows; Supplementary Material 4). As a backstop we also fitted constrained models that share the landscape parameters (σ_c, β) across durations, leaving only the sensory channel and lapse free per duration. Constraining the landscape to be duration-independent costs $\Delta\text{AIC} = 3529$ (sharing both; $\Delta\text{BIC} = 3458$, so the penalty is not a parameter-count artefact), with the cost concentrated in σ_c (sharing only σ_c costs 3398, only β costs 990). The landscape therefore changes with exposure, mostly through σ_c: the wells sharpen as the sensory evidence improves. We read this as a change in the resolution of one mechanism, not a change of mechanism.

Table S3. Trial-level parameter recovery of the winning model. Identity R^2 and Spearman ρ between generating and recovered values across the 250 synthetic datasets. Scale parameters are compared on the log scale.

Quantity	R^2	Spearman ρ
σ_c (well scale)	0.99	1.00
β (width exponent)	0.89	0.95
s_0 (sensory scale)	0.95	0.96
λ_0 (sensory exponent)	0.83	0.94
ℓ (lapse)	0.97	0.98
n_{crit} (derived)	1.00	1.00
σ_3 (well width at $n = 3$)	0.98	1.00
$\sigma(n = 5)$ (sensory sd)	0.99	1.00

A frequency prior at the readout does not help. In the winning model the number frequency f_n enters only through the well width. We also tested adding a frequency prior at the readout stage ($p(n \mid x^*) \propto f_n \exp g_n(x^*)$), a zero-parameter change. It does not robustly improve the fit (summed $\Delta\text{AIC} = -125$ but inconsistent in sign across durations, helping at 80–160 ms and hurting at 40, 320, and 640 ms), so we retain the flat readout and let the frequency dependence live entirely in the well width.

Joint channel. Co-fitting the response and response-time channels together, the winning attractor with the dynamics-based RT link remains best, ahead of its static twin by $\Delta\text{AIC} = 1006$ (clean-up still helps once RT is included) and of the Weber/ANS code by 4155. Within the winner, the convergence-time RT link improves the joint fit over the static entropy proxy, as reported in the main text.

Supplementary Material 3: Parameter recovery

To confirm that the fitted quantities are identifiable at the trial level, we ran a parametric recovery: for each duration, 50 synthetic datasets were generated from parameters jittered around the fitted solution, sampled with the empirical per-numerosity trial counts and the lapse channel, and refitted with the exact headline pipeline (250 refits total). Table S3 reports the recovery of each parameter and of the derived quantities the paper interprets; Figure S1 shows the scatter and the error-correlation structure.

The width exponent β and, especially, the derived resolution limit n_{crit} (the two quantities that carry the paper's argument) are recovered essentially perfectly ($R^2 = 0.89$ and 0.996). The only strong trade-off in the error-correlation matrix is between the two sensory parameters s_0 and λ_0 ($\rho = -0.97$); accordingly we never interpret them separately, only through the combined sensory-noise profile $\sigma(n) = \sqrt{s_{00}^2 + s_0 n^{\lambda_0}}$, which is itself recovered accurately. All landscape quantities (σ_c, β, n_{crit}) are cleanly identified.

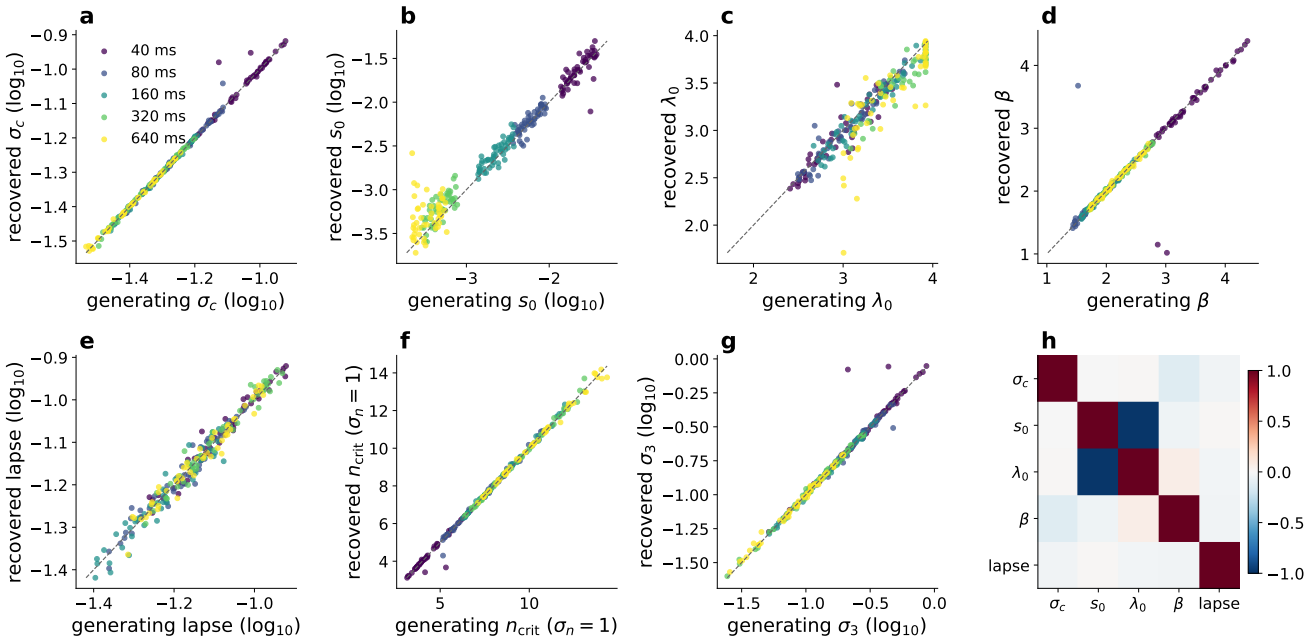


Figure S1. Parameter recovery of the winning model. Recovered versus generating values for the five fitted parameters and the derived n_{crit} and σ_3, coloured by exposure duration; dashed lines are identity. The bottom-right panel is the Spearman correlation matrix of recovery errors, showing the single strong s_0–λ_0 trade-off.

Supplementary Material 4: The resolution limit n_{crit}

The critical numerosity is defined by adjacent-well discriminability $d' = 1$, i.e. $\sigma_n = 1$ (well standard deviation equal to unit spacing) (Macmillan and Creelman 2004). Solving $\sigma_n = (\sigma_c / f_n)^{\beta} = 1$ for the fitted parameters gives $n_{\mathrm{crit}} \approx 4.6,\ 7.5,\ 8.7,\ 9.2,\ 10.0$ for 40–640 ms. The crossing lies inside the fitted range at every duration and grows monotonically with exposure (main-text Fig. 2c).

The landscape is intrinsically small-range. Because the crossing falls inside the fitted window, one might worry that the large-n tail distorts the landscape. It does not: refitting on $n \leq 10$ alone returns nearly the same width exponent and n_{crit} as the full-range fit for 80–640 ms (e.g. $\beta = 1.88$ vs. 1.81 at 80 ms, 2.09 vs. 2.19 at 320 ms; n_{crit} within 0.3 at all four), and only the noise-dominated 40 ms flash shifts (Fig. S2). Within $n \leq 10$ the winning model still wins, ahead of its static twin ($\Delta\mathrm{AIC} = 172$) and the Weber/ANS code (2043). The exact code is thus a property of the small-number range, not an artefact of fitting the tail.

Cross-channel convergence with the response-time plateau. Within the exact range ($n < n_{\mathrm{crit}}$) the model tracks the geometric-mean response time closely; above it the monotone deliberation-time prediction keeps rising while the empirical RT plateaus, because participants have left the exact regime and are guessing approximately (Fig. S3). The onset of the empirical RT plateau coincides with n_{crit}, which is derived independently from the accuracy channel, a convergence of two channels on the same boundary. We therefore fit the latent-to-seconds RT link on all n and report the within-domain agreement; truncating the fit to $n < n_{\mathrm{crit}}$ is counter-productive, since at short durations the cut falls just before saturation and the truncated fit overshoots.

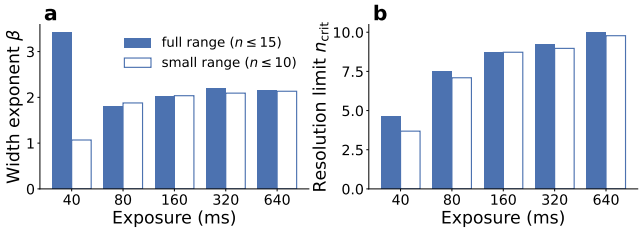


Figure S2. Small-range refit. The fitted width exponent β (**a**) and the resolution limit n_{crit} (**b**), from the full-range fit ($n \leq 15$, filled) and from a refit on the small range ($n \leq 10$, open). Both are essentially unchanged for 80–640 ms; only the noise-dominated 40 ms flash shifts. The discrete, exact behaviour is intrinsic to the small-number range.

The discrete signature in exact accuracy. The four-parameter landscape is smooth, yet the predicted proportion of exact responses shows a small shoulder in the subitizing range. This is not extra fitted flexibility: it is the discrete clean-up basins on the fixed integer lattice, and it vanishes when the clean-up is switched off (Fig. S4). The wiggle is thus a signature of the mechanism rather than of additional degrees of freedom.

The effective Weber fraction. We computed the coefficient of variation (CoV) of the model's fitted response distribution — $(1-\ell)\,p(r \mid n) + \ell/R_{\max}$, the model exactly as fitted, with the lapse included — per numerosity and exposure, restricted to the in-support range so that it is the like-for-like counterpart of the empirical estimates, and overlaid the empirical CoV (Fig. S5). The model tracks the data across the whole range at every exposure. The CoV is elevated at the smallest numerosities (where the ratio $\mathrm{sd}/\mathrm{mean}$ is inflated by the small denominator, in model and data alike) and settles to a roughly constant plateau across the estimation range ($n \geq n_{\mathrm{crit}}$). A power-law fit $w_{\mathrm{eff}} \propto n^{\gamma}$ over

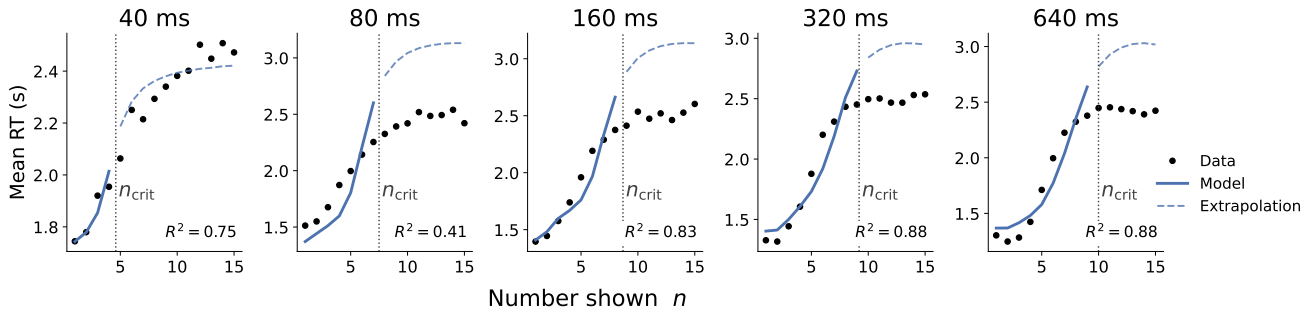


Figure S3. Response time within and beyond the exact range. Predicted geometric-mean RT (solid, $n < n_{\mathrm{crit}}$; dashed, extrapolated above) against the data. Within the exact range the model tracks RT; above n_{crit} it over-predicts, the signature of the switch to approximate responding. The RT plateau onset coincides with the independently derived n_{crit}.

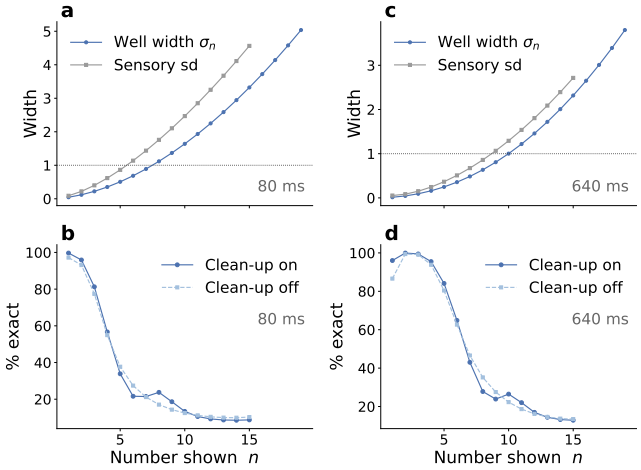


Figure S4. The exact-accuracy shoulder is a clean-up signature. Top: the fitted well-width and sensory-noise profiles are smooth. Bottom: the proportion of exact responses shows a discrete shoulder with clean-up on (solid) that disappears with clean-up off (dashed), the discrete basins on the integer lattice, not extra parameters.

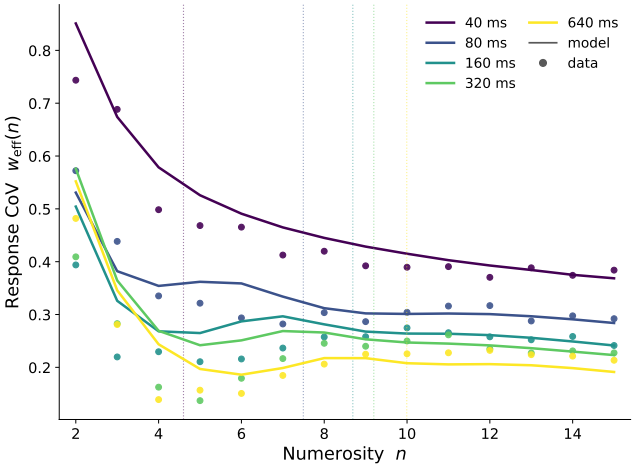


Figure S5. The effective Weber fraction. Coefficient of variation of the response distribution $w_{\mathrm{eff}}(n)$ per numerosity, for each exposure duration; solid lines are the model (the fitted response distribution, lapse included, restricted to in-support and renormalised), dots the data (in-support responses), and dotted verticals mark n_{crit}. The model tracks the data across the whole range; across the estimation range ($n \geq n_{\mathrm{crit}}$) the CoV settles to a roughly constant plateau, approximately scalar variability (Weber's law). The 40 ms flash is noise-dominated and has no exact range.

the estimation range gives a pooled exponent $\gamma \approx -0.2$ for both the model (-0.21) and the data (-0.17): approximately scalar variability, close to the Weber value $\gamma = 0$ and far from the $1/\sqrt{n}$ decline ($\gamma = -0.5$) that a Poisson magnitude code would predict. The model thus reproduces the measured variability, and its slope, across the estimation range — one landscape accounting for numerosity variability in both regimes.

Supplementary Material 5: Exploratory reanalysis of developmental dyscalculia

As an exploratory extension we applied the winning model to the exact-enumeration data of Decarli et al. (2020): accuracy and response time per subject and numerosity for 32 children with developmental dyscalculia (DD) and 32 age-matched controls, using numerosities $n = 1$–7 (dropping $n = 8$ exactly as the original authors do, for the guessing end-effect they report). The winning model fits both groups well and independently prefers a super-linear width ($\beta \approx 1.5$–1.6), a cross-dataset replication, with a re-derived $n_{\mathrm{crit}}(\sigma_n = 1) \approx 8$, a plausible resolution limit for exact enumeration. The argument has three tiers, strongest-anchored first (Fig. S6).

Model-agnostic deficit. DD are $\approx 10\%$ less accurate at $n = 4$–6 (Cohen's $d = -0.62$) and ≈ 0.3 s slower at $n = 4$–7 ($d = +0.61$), with the difference concentrated at $n = 4$ (Mann–Whitney $p = 0.001$), the subitizing edge, not a uniform slowdown. This needs no model.

The deficit is resolution-shaped, not noise-shaped. With an exhaustive per-subject grid search and a fixed common lapse, a single-mechanism *resolution* model (well merging, sensory noise off) fits each subject's accuracy curve far better than a single-mechanism *noise* model (well width off): $64/64$ subjects, mean SSE 0.010 vs. 0.079, Wilcoxon $p = 3.5 \times 10^{-12}$. The exact-enumeration decline is a discrete-resolution phenomenon, which both validates the attractor framework on independent data and says where the group effect must live.

The group difference localises to resolution. Per-subject, the resolution limit is shorter in DD ($n_{\mathrm{crit}} = 8.2$ vs. 9.1 for controls; Mann–Whitney $p = 0.004$ with the RT channel included). Sensory noise shows only a weak trend ($p = 0.07$)

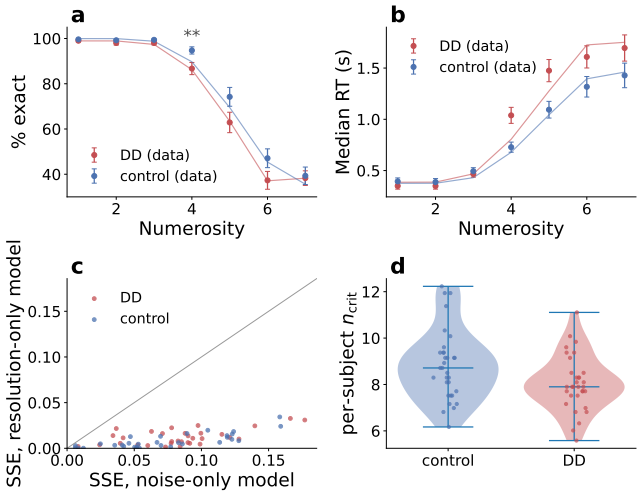


Figure S6. Developmental dyscalculia as a shorter exact number code. (a) Exact accuracy: DD are less accurate, concentrated at $n = 4$ ($**$). (b) Response time: DD are slower in the transition band. (c) A resolution model fits each subject's accuracy better than a noise model ($64/64$; below the diagonal). (d) The per-subject resolution limit n_{crit} is shorter in DD.

and fits worse. The standardised n_{crit} effect ($d = -0.65$) matches the raw accuracy effect, so the modest raw shift of ≈ 0.9 numerosities is a units effect, not a small one.

The reading for the paper is that developmental dyscalculia is, on this model, *not* a noisier perception of number but a coarser, shorter exact code: the discrete resolution runs out earlier, right at the subitizing edge. This is a specific, mechanistic, falsifiable claim, though much of the statistical power comes from analysing subjects individually rather than at the group level, not from the model alone. It is complementary to the account of Dolfi et al. (2024), who locate the dyscalculic deficit in a weaker number sense within an image-computable network: both find the deficit in the number code, and coupling our attractor to such a front end (the present model sets the front end aside) is the natural way to ask whether a resolution deficit and a sensory-noise deficit can be told apart in the same model. The analysis remains exploratory: the data are coarse (8 trials per cell), the resolution-versus-noise localisation is a lean supported by the decisive per-subject model comparison rather than a full identification, and a small sensory-noise component may coexist.

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